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Algae extracts as bio-effectors for improved growth, mineral nutrition and stress resilience of Durum wheat



Plants effectors-a novel product category for agriculture

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Introduction: Modern Challenges for Agriculture

Reduce environmental impacts & mitigate climate change

Global N fertiliser use 195 Mio t. year⁻¹, < 40% use efficiency
(Yu et al., 2023)

Global P fertiliser use 20.5 Mio t. year⁻¹, <10% use efficiency
(McDovell et al., 2023)

Global pesticide use 7.2 Mio t. year⁻¹ (FAO, 2022)

→ Bio-effectors as alternative to agrochemicals ?

→ Enhance plant growth & stress resilience (Khan et al., 2009).

→ Improved nutrient use efficiency (Craigie, 2010)





Open Climate Solutions

Linking regenerative aquaculture with regenerative agriculture

... harnessing the power of seaweed to regenerate oceans, soils and communities

bringing it all together into a climate solution and helping farmers



REGENERATIVE. SUSTAINABLE.



Gut Wilhelmsthal

... modern farm ... efficient
sustainable farming methods and
environmentally conscious action



Healthy soil = healthy plants = healthy animal = healthy person!

Problem statement

Sustainability

Impact of Fertilizers & Pesticides

Nutrient Use Efficiency

Soil Health

Environmental Problems

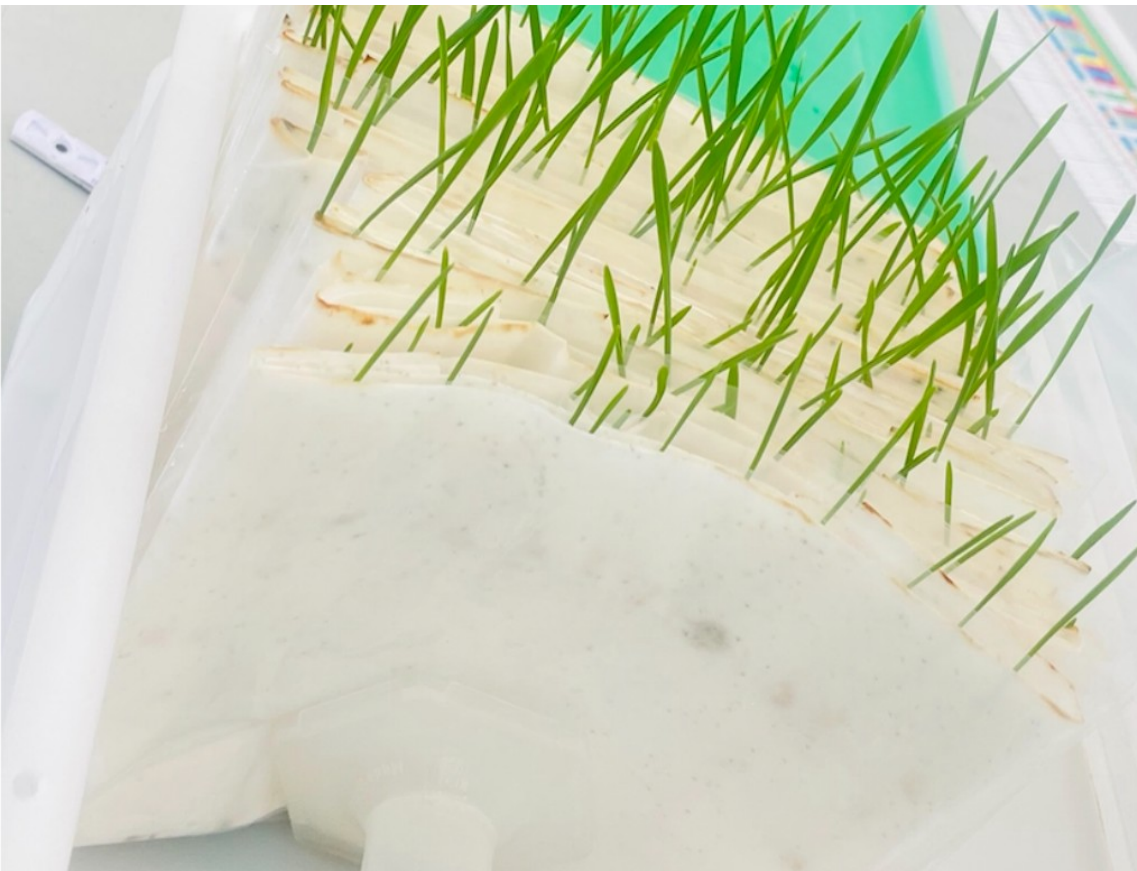


Research Objectives

The efficacy of seaweed extracts based bioeffectors:

- 0 Adaptation to common fertilization strategies
- 0 Nutrient use efficiency
- 0 The contribution to Eu farm-to-fork strategy

Materials and Methods



Screening test

- Durum wheat seeds were tested for germination.
- Algae extracts doses were 0.0, 0.5, 2.5, and 5%.
- Filter paper soaked with 50 ml algae extracts.
- Germination observed at room temperature (24°C).
- Root length and number root tips measured.
- Coleoptile length recorded for each seed.

Experimental design

- Pot experiment in Greenhouse at University of Hohenheim
- The spring durum Wheat var Beladur was used
- 1 Variey×8 treatments ×5 replications = 40



SPA+BEM	AMLLK	5SPB24P5	Ukelp	0.5SPB24P5	SPAP4	Ctrl	5SPB24P5	AMLLK	SPAP4	SWP4	
0.5SPB24P5	AMLLK	SPAP4	Ukelp	SWP4	SPA+BEM	Ukelp	SWP4	SWP4	SPA+BEM	0.5SPB24P5	SPAP4
Ukelp	Ctrl	Ctrl	SPA+BEM	Ctrl	5SPB24P5	Ctrl	5SPB24P5	0.5SPB24P5	5SPB24P5	AMLLK	

Bio-effector	Species	Production	Current Status
AMLLK, Atlantik Mariculture Limited, UK	Seaweed algae (S. latissma)	fermentation	Commercial product
SWP4, Open Climate Solutions		Wet milling	innovative product
SPA+BEM, Open Climate Solutions	Freshwater algae (Charophyta)	Fermentation	innovative product
SPB24P4, Open Climate Solutions		Wet milling	innovative product
SPAP4, Open Climate Solutions		Wet milling	innovative product
Ukelp, Abundigro, Netherlands; Liquid KelpTM,	E. Maxima seaweed algae	Pure cold pressed	Commercial product 10

Soil substrate

- Topsoil of grassland
- It is from campus garden
- It is a Parabraunerde
- it has high water holding capacity
- And it for is rich in Ca, Mg, and K.

Soil features	Values/information
Parent rock	loes
Horizon	humic topsoil layer under grassland
soil pH (CaCl ₂)	7.4
Extractable P, K, Mg	P (CAL): 36 mg kg⁻¹ soil dry matter content class C (adequate) K (CAL): 96 mg kg⁻¹ soil dry matter content class C (adequate) Mg (CaCl₂): 136 mg kg⁻¹ soil dry matter content class C (adequate) K:Mg ratio 3:1 for optimal acquisition by the plant, therefore no Mg fertilizer was applied
Texture	silty clay loam
Maimum water holding capacity (WHC max)	33,2 g water 100 g ⁻¹ soil dry matter

Nutrient solution preparation

- Potassium sulfate solution prepared to deliver potassium.
- Ammonium phosphate solution provides phosphorus and nitrogen.
- Calcium nitrate fulfills remaining nitrogen requirement in solution.

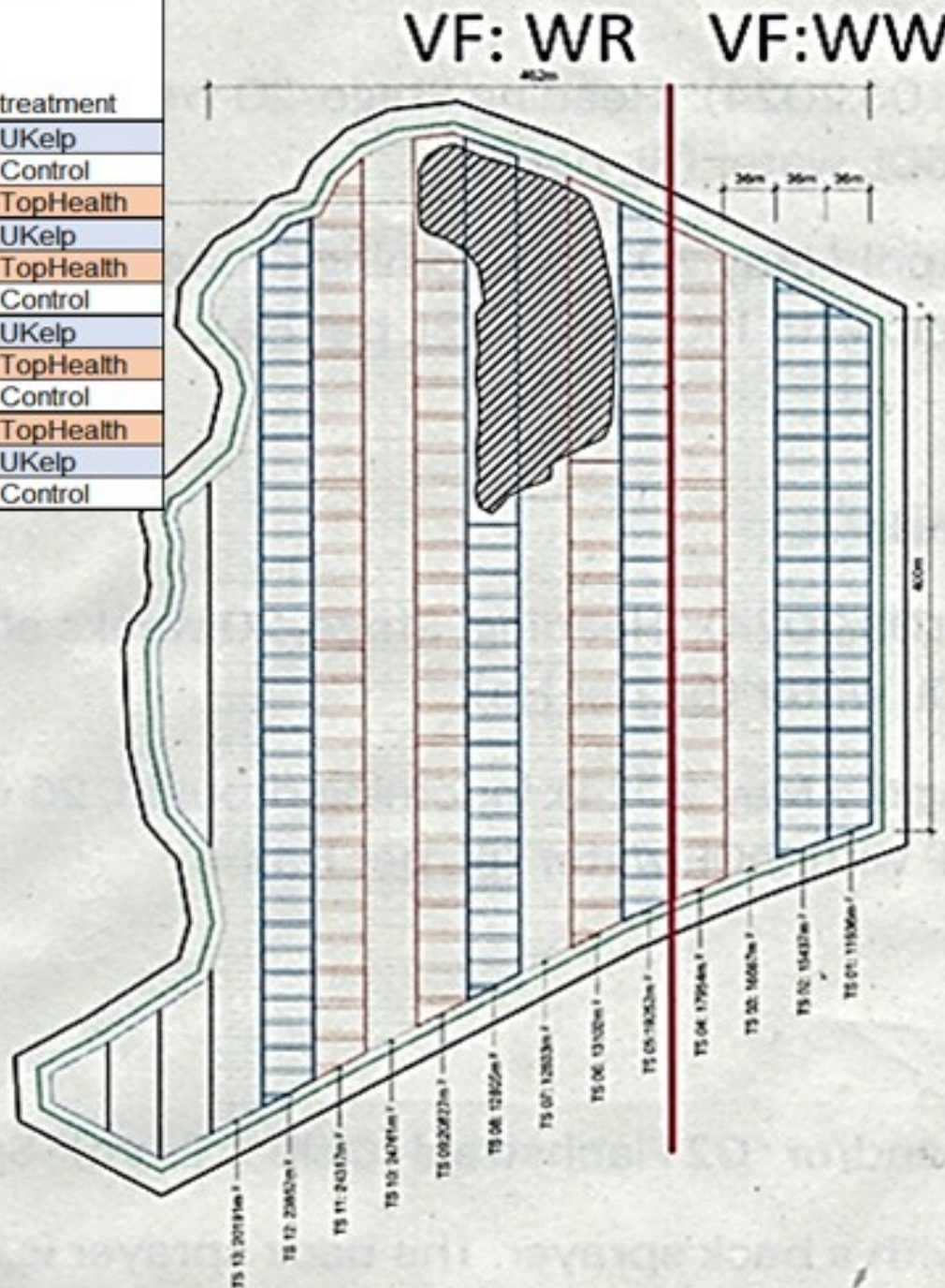
Target values [mg kg ⁻¹ SoilDM]	
N	60
P	80
K	150

Planting, Watering and Algae extracts application

- The Mitscherlich pots (314.14 cm²)
- Field sowing density of 22 viable seeds/pot
- Planting date 26th February 2024
- Covered with quartz sand N°7
- Watering kept the moisture between 17-23%.
- Drought phase introduced of 23 days from 70th DAP
- Algae extracts sprayed twice on 54 & 84 DAP
- The doses were determined by screening test



block	plot number (trial site)	treatment
I	2	UKelp
I	3	Control
I	4	TopHealth
II	5	UKelp
II	6	TopHealth
II	7	Control
III	8	UKelp
III	9	TopHealth
III	10	Control
IV	11	TopHealth
IV	12	UKelp
IV	13	Control



Field experiment

- it was a randomized block design
- Gut Wilhelmsthal farm, Saxony-Anhalt
- 26.44 hectares of sandy loam soil.
- Winter wheat (var. Emerick, KWS)
- Sown on 22nd September, 2023.
- 3 variants × 4 replication = 12 plots

Application

A red tractor is pulling a large agricultural sprayer across a green field. The sprayer has a long boom with multiple nozzles, and a fine mist of spray is visible being applied to the crops. The background shows a clear sky and distant hills.

- Three variants were tested: Control, Ukelp and TopHealth

- **Products and Application:**

- **Control** (8.36 ha)
- **Ukelp:** Applied at 3 L/ha (9.37 ha),
- **TopHealth:** Applied at 5 L/ha (8.71 ha).
- Two Applications: 20 WAP (tillering phase,), and in May.



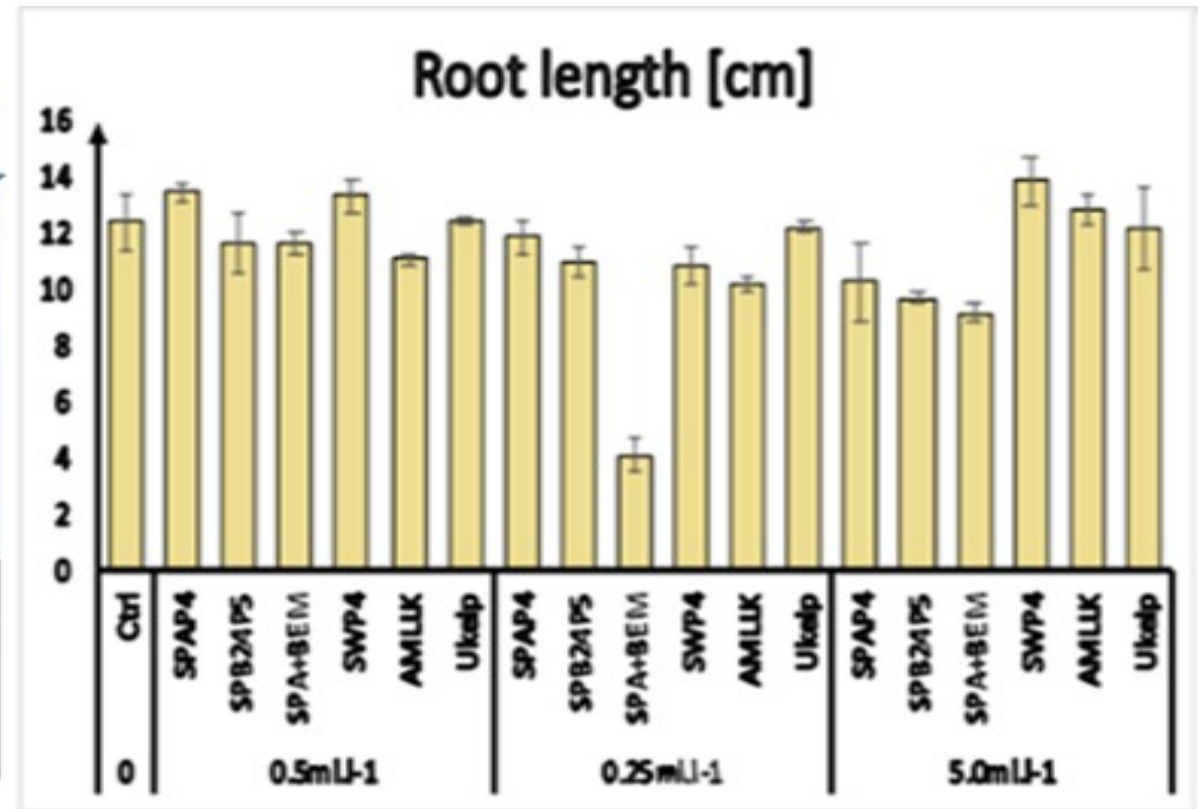
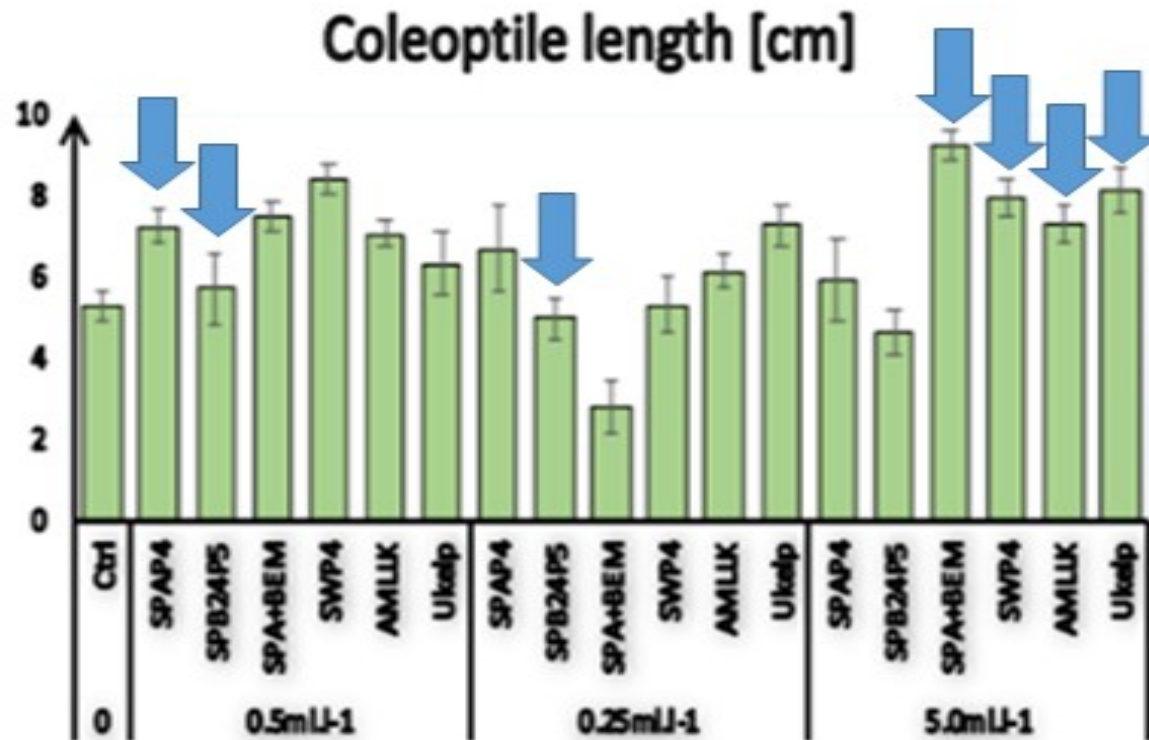
Data collection and Analysis

- Screen test and growth parameters
- Yield Parameters and mineral Nutrient analysis
- ANOVA determined significant differences (F-test)
- Mutiple T-test for mean comparison (Mudra, 1958).
- SAS and R software were used.

Results

Screening test

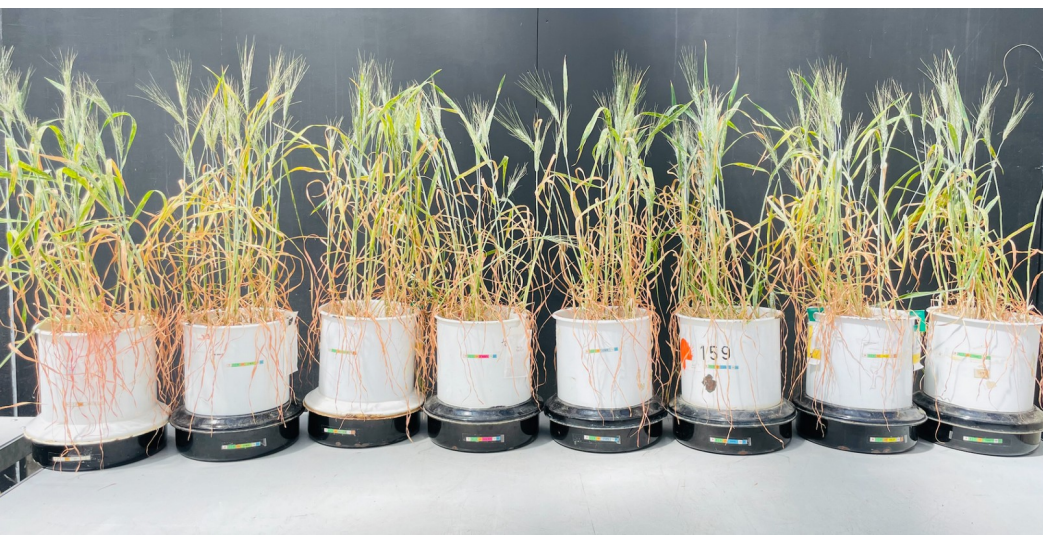
- There was notable effects on coleoptile length
- No influence on root length or the number of root tips under the experimental conditions.



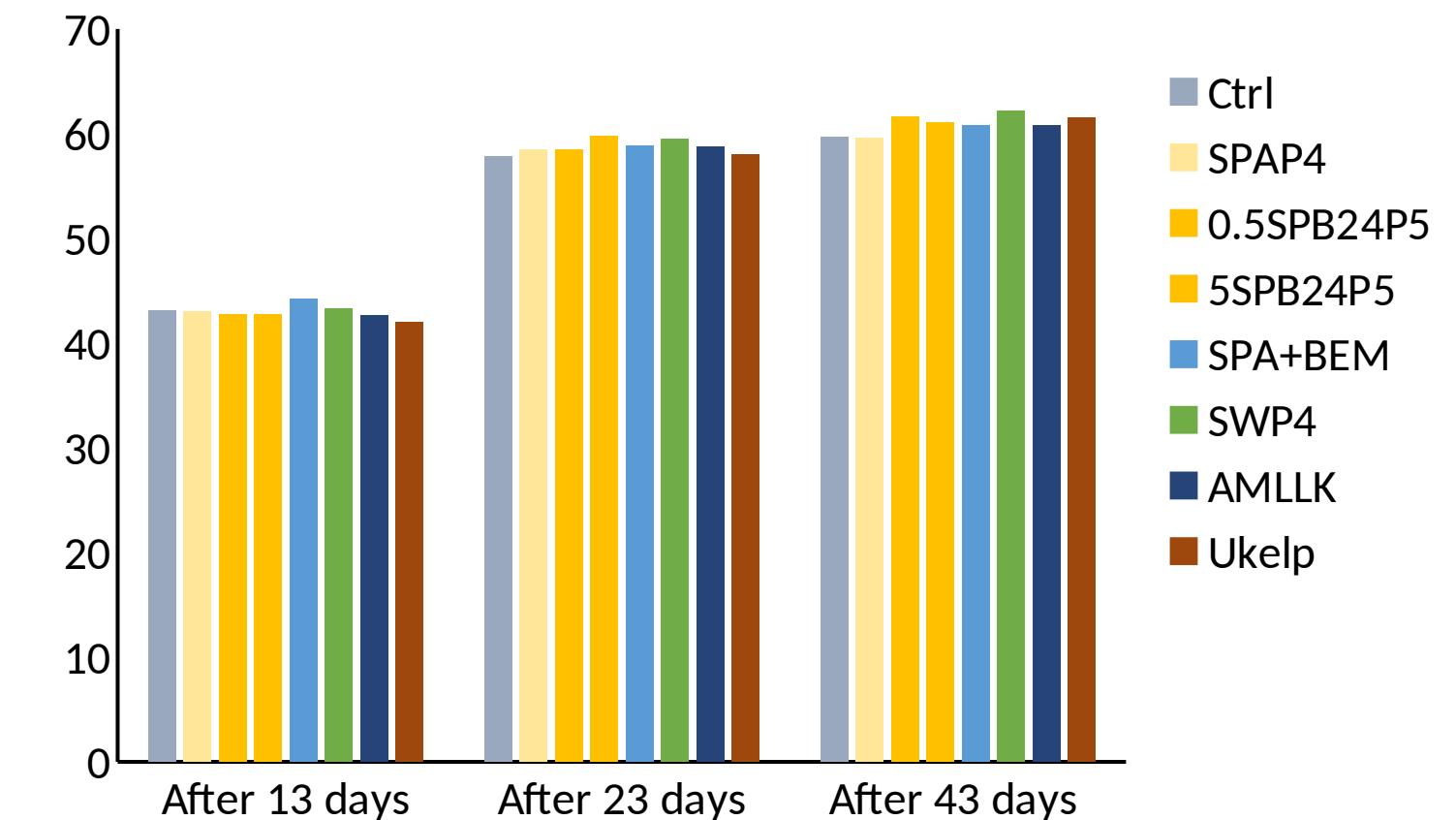
The blue arrows indicate which concentrations were collected for the application in the pot experiment.

Plant height

- Bio-effector treatments showed no significant height differences.
- Plant height increased similarly across all treatments.



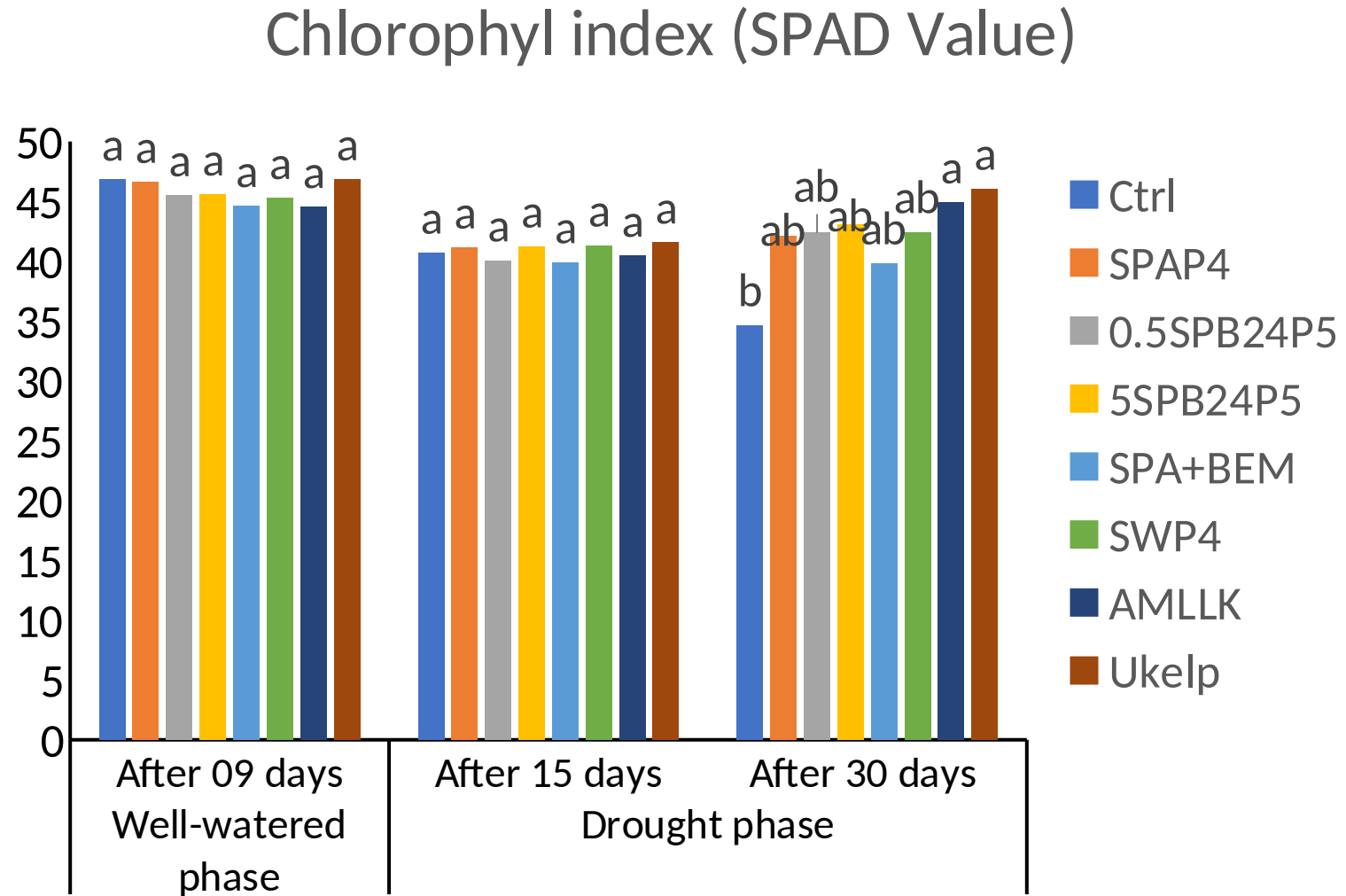
Plant height (cm)



Plant height at 13, 23, 43 days after 1st application of seaweed extracts

Chlorophyll index

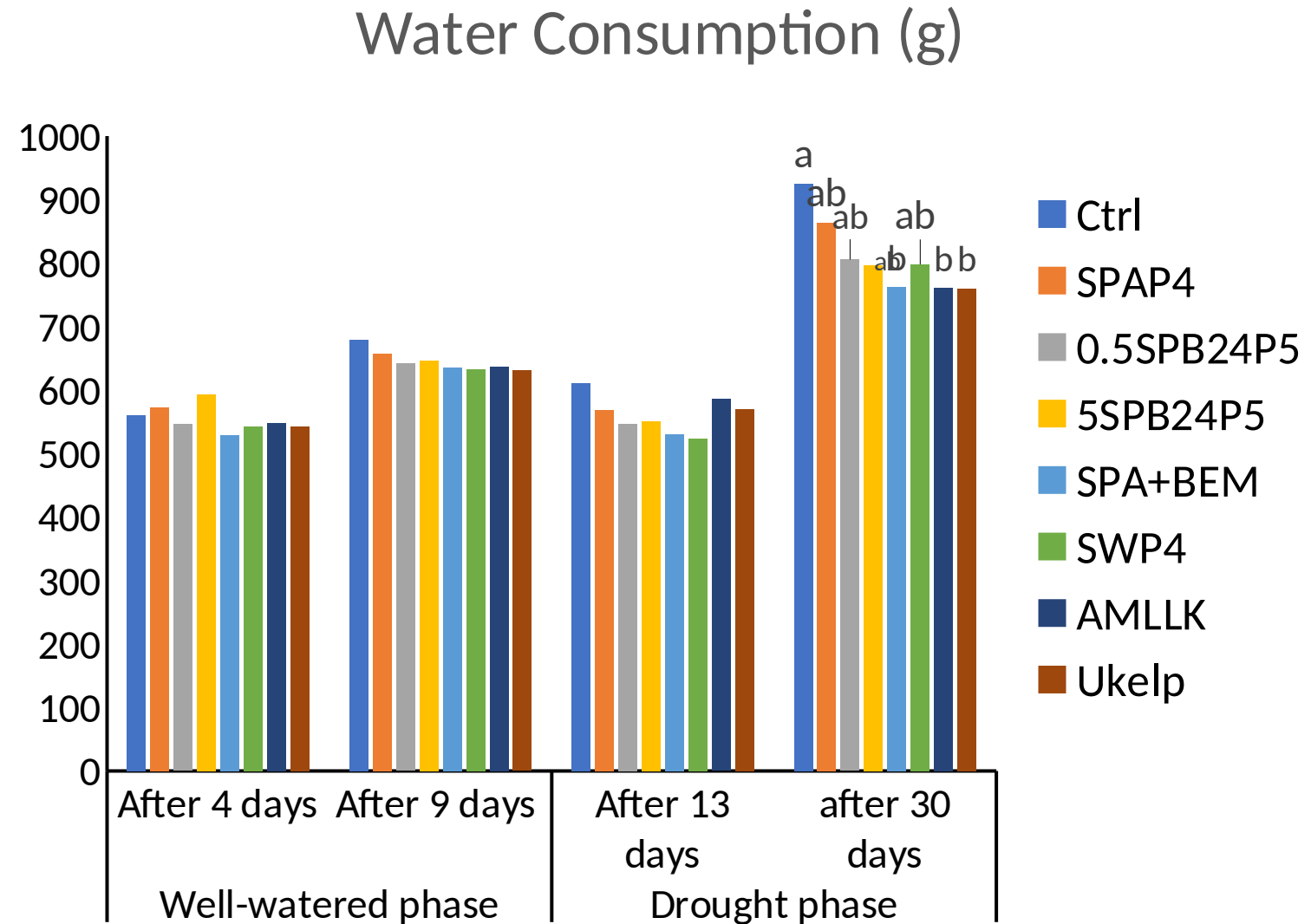
- No significant SPAD value differences under normal conditions.
- Ukelp and AMLLK showed significant higher SPAD values under drought phase.



Effect of bio-effectors application on leaf chlorophyll index of Durum wheat in a pot experiment. The treatments with the same letters are not significantly different ($P \leq 0.05$).

Water Consumption

- No significant differences in water consumption during watering phase.
- Control treatment consumed more water than bio-effectors.
- SPA+BEM, AMLLK, and Ukelp showed lowest consumption.



The Water consumption at 4, 9, 13, 30 days after 1st seaweed extracts application.
The treatments with same letters are not significantly different ($p \leq 0.05$).

Dry matter biomass

- Bio-effector treatments generally showed similar results to control.
- 5 ml/l SPB24P5 stimulated the highest root biomass.
- AMLLK treatment produced the lowest root biomass.
- 0.5 ml/l SPB24P5 resulted in the highest ear biomass.

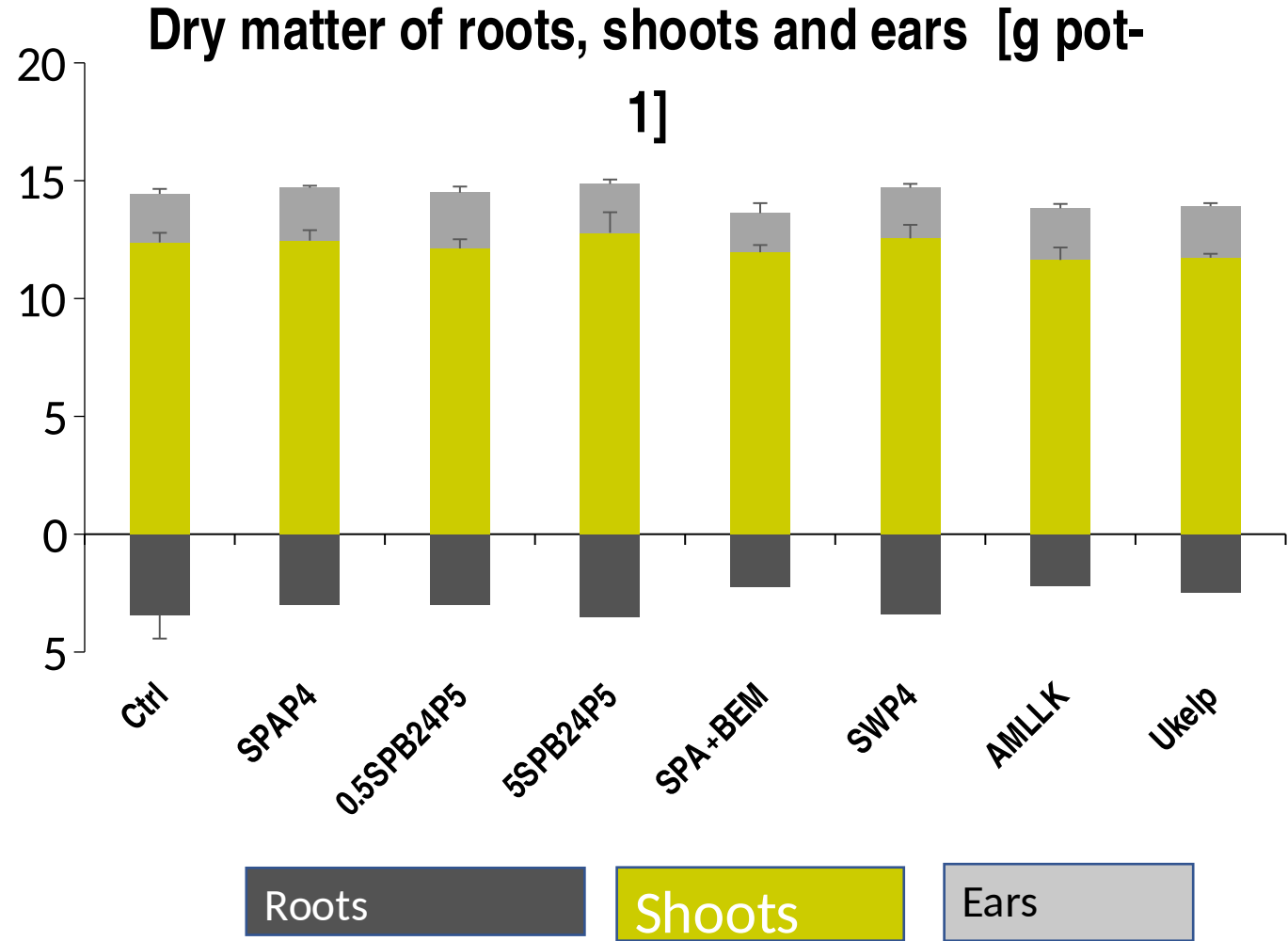
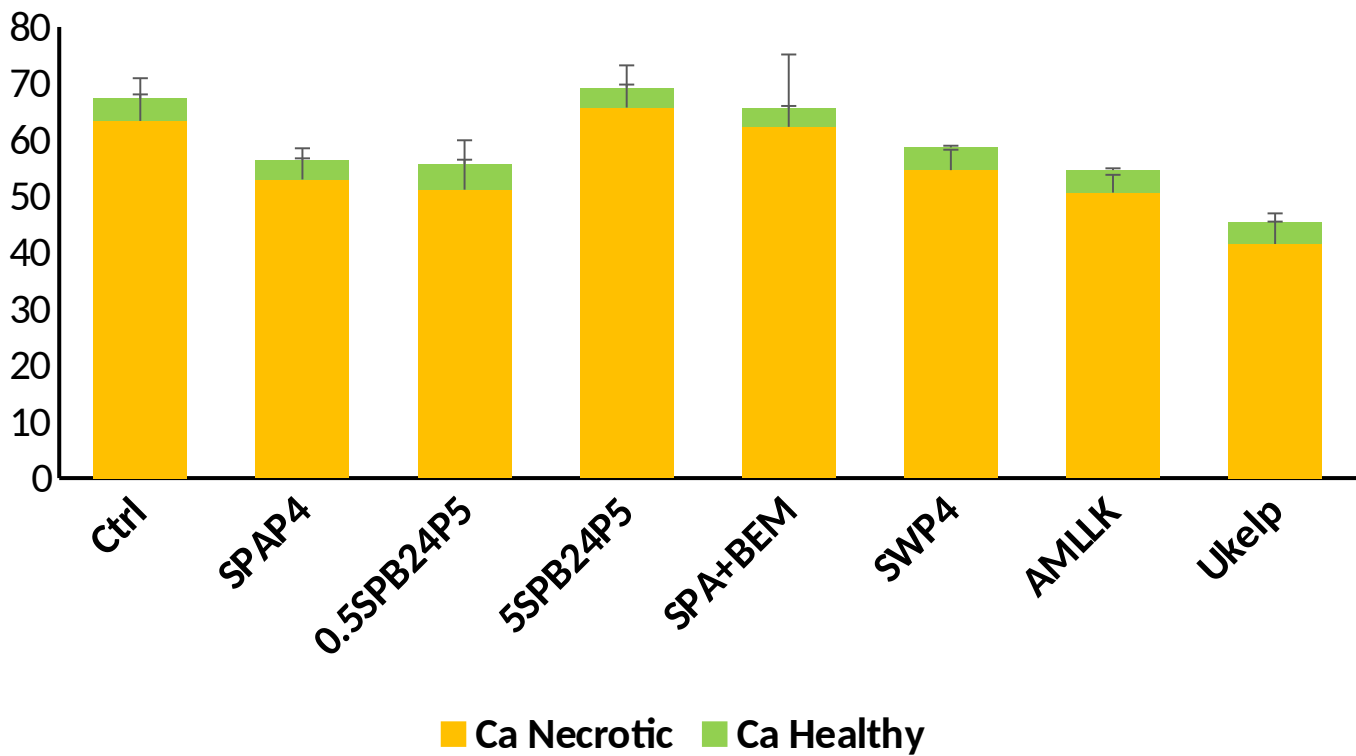


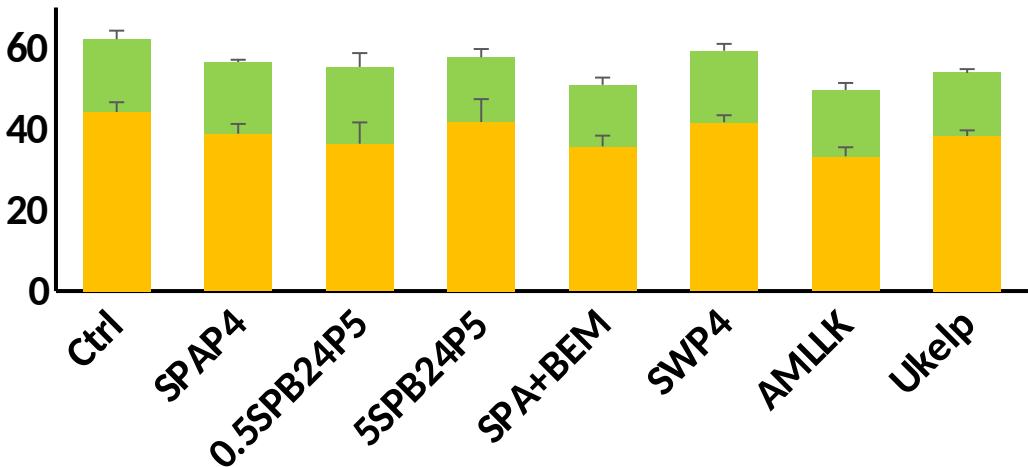
Figure: Root, shoot and ear dry biomass of pot experiment harvest 10 months and 20 days after sowing. The means were not significantly different ($p > 0.05$).

Nutrient analysis for whole biomass

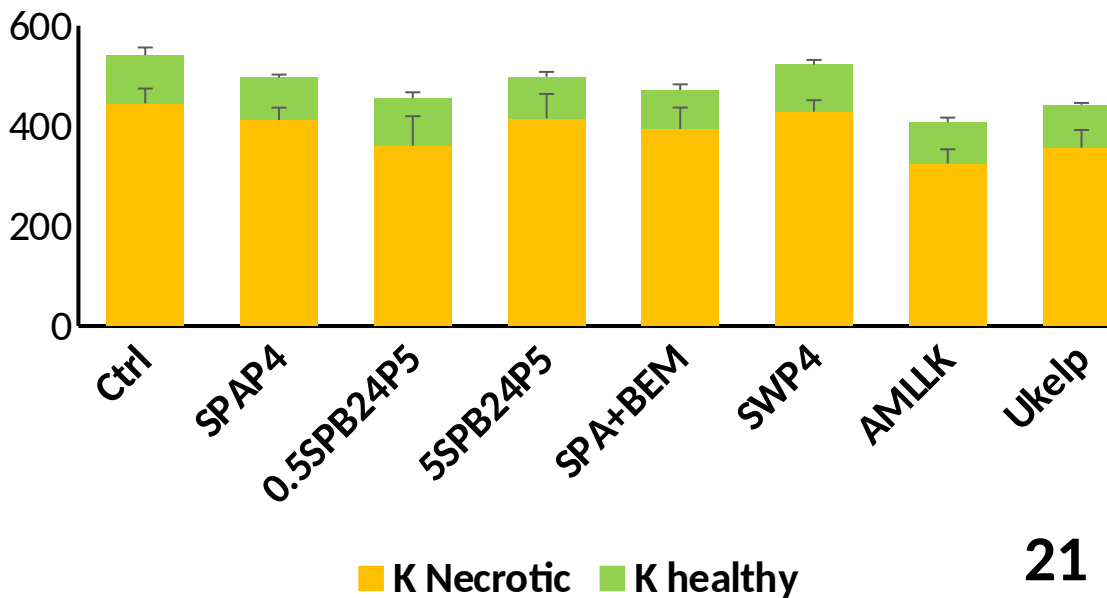
Calcium content [mg.pot-1]

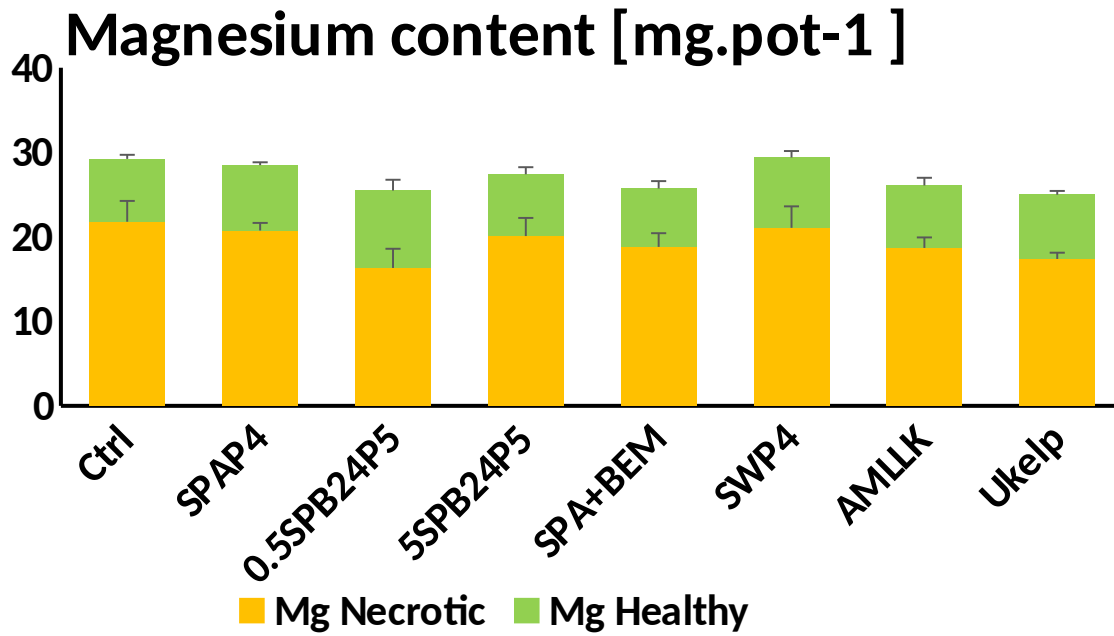


Phosphorous content [mg.pot-1]

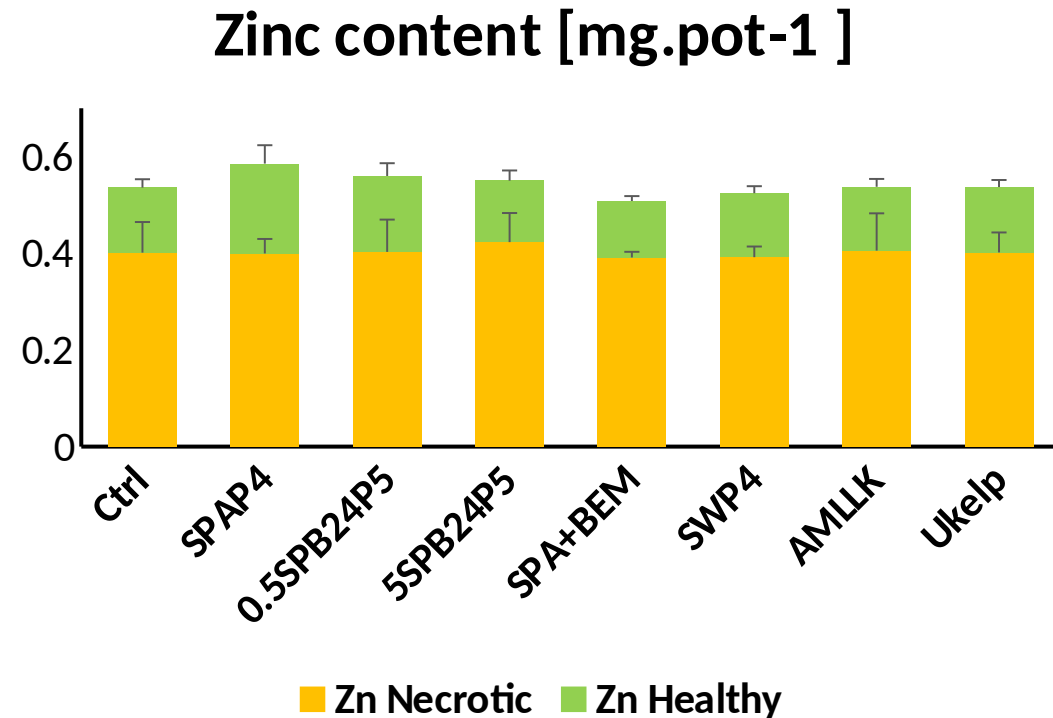


Potassium content [mg.pot-1]

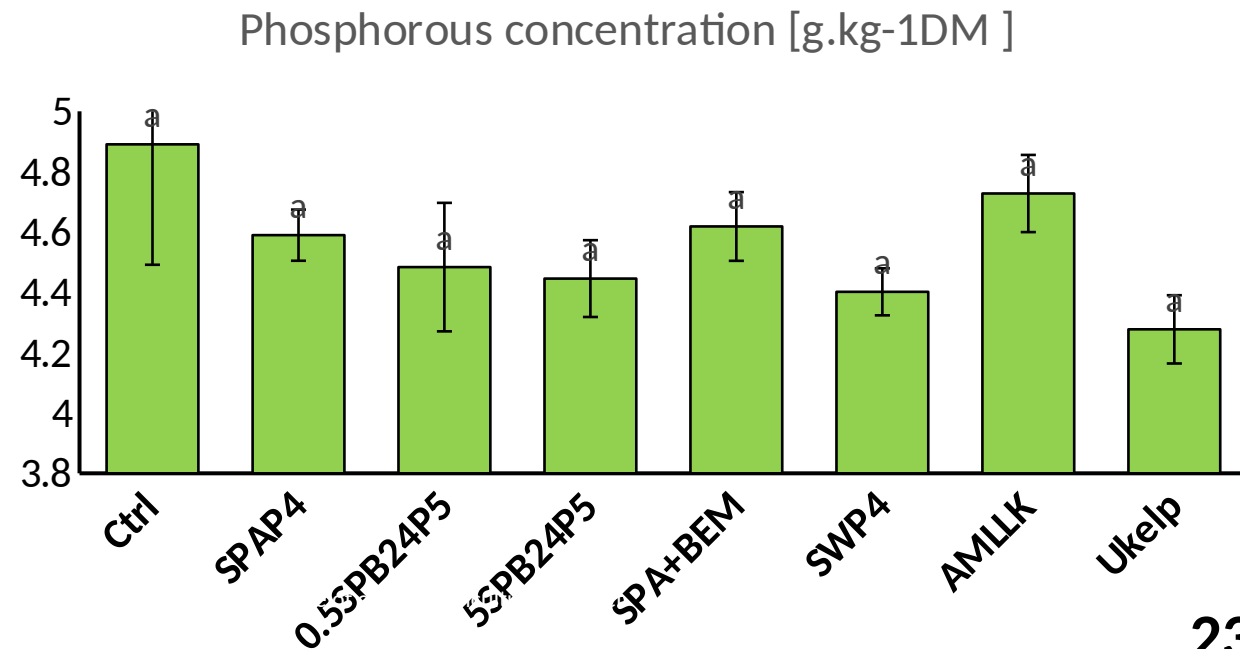
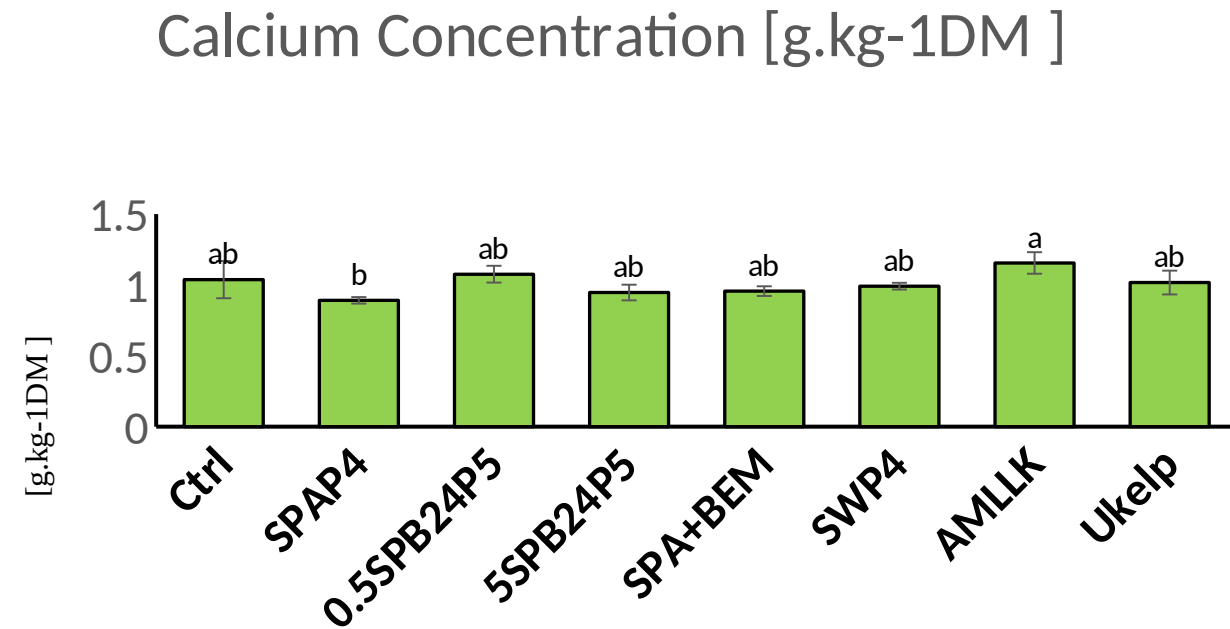
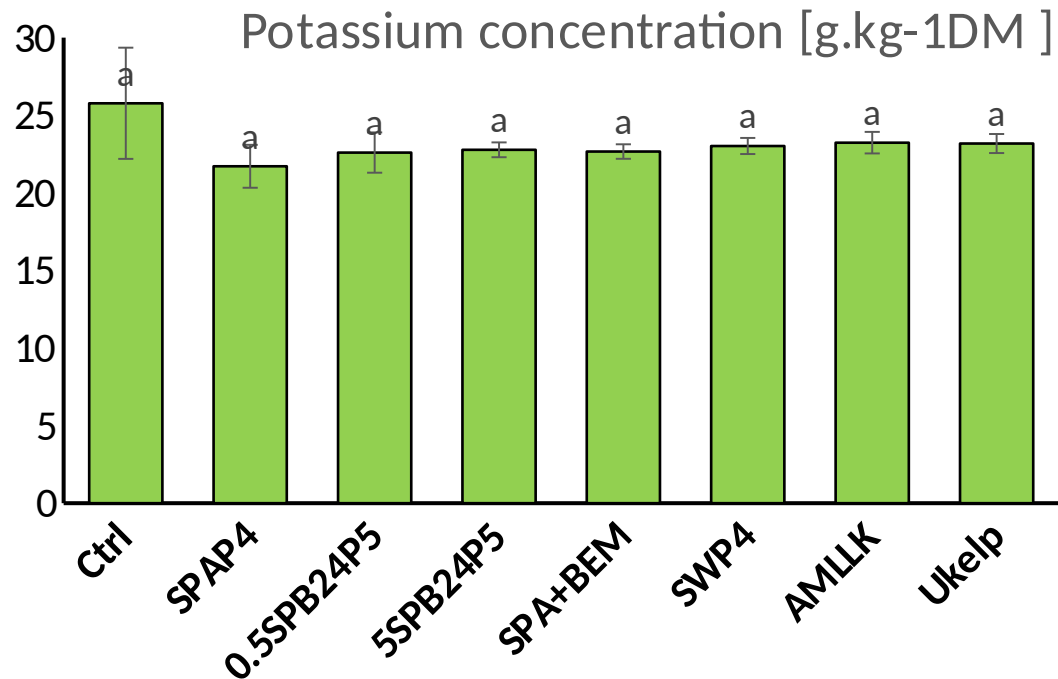




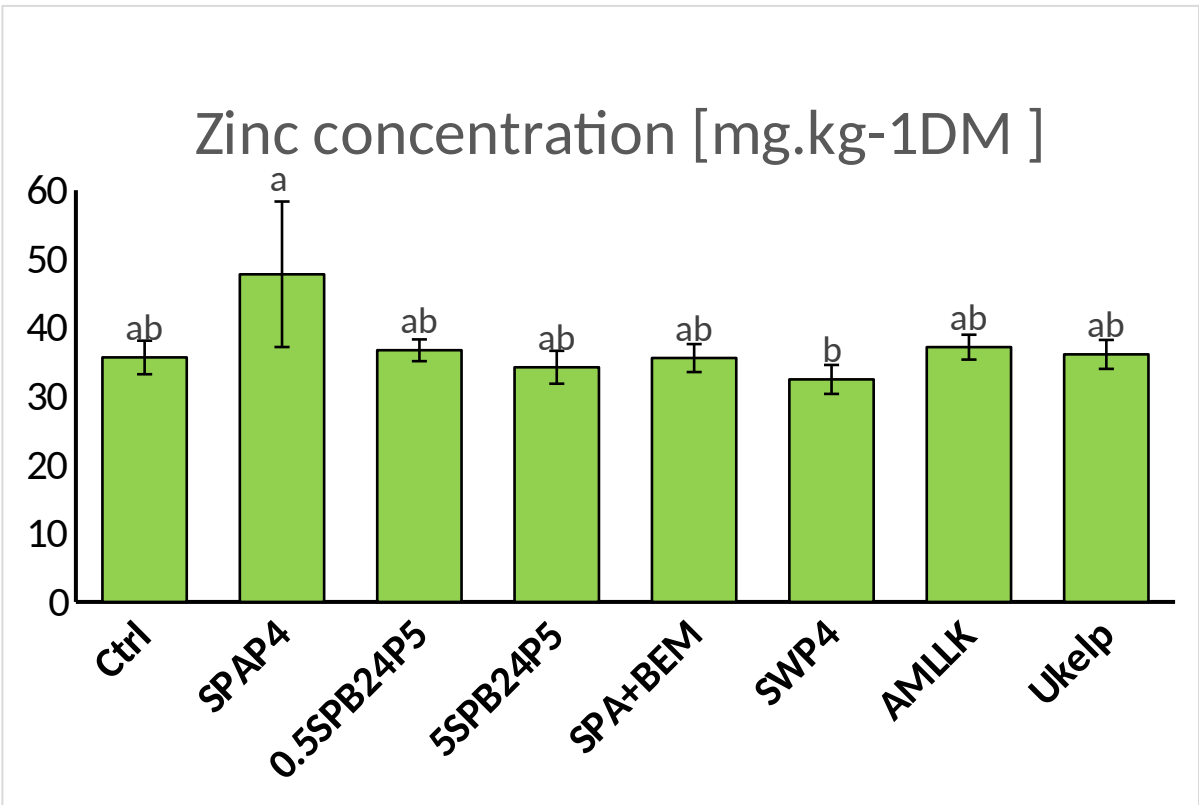
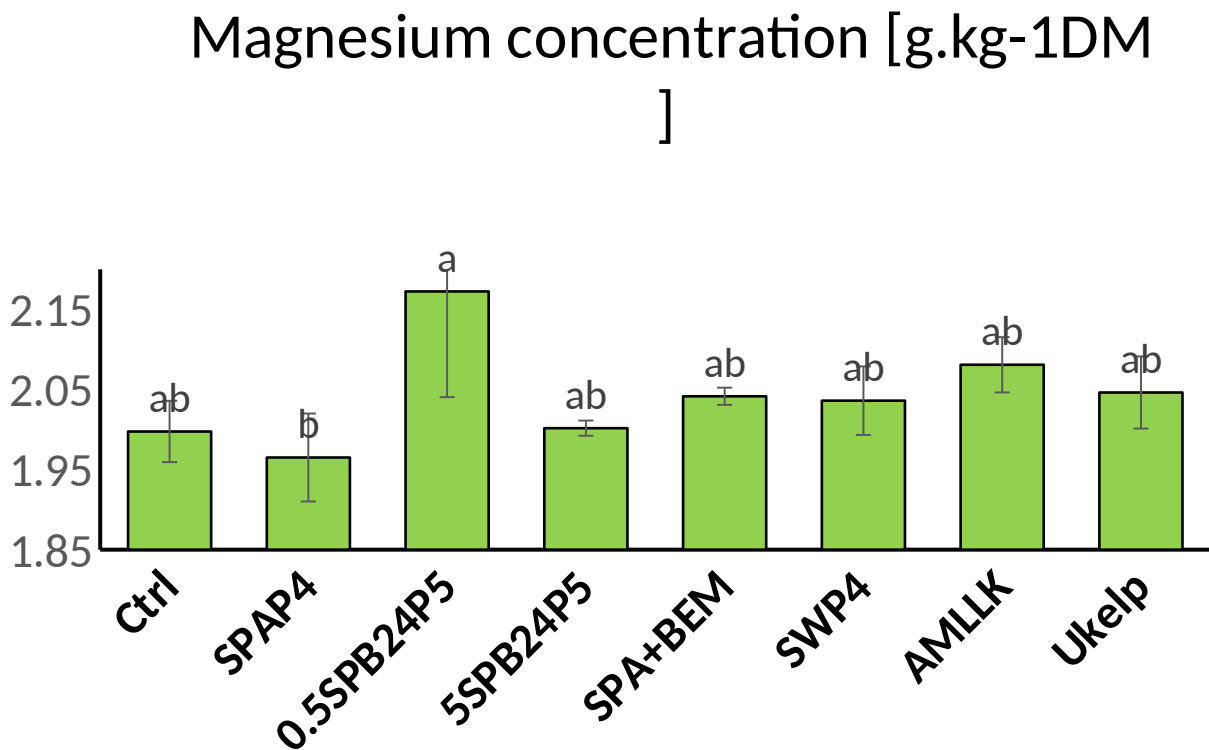
- The means were not significantly different ($p > 0.05$).
- There was no significant effect on mineral content as whole biomass.
- Control had the highest potassium and phosphorus in necrotic parts.
- 5 ml/l SPB24P5 showed highest calcium and zinc levels.
- SPAP4 recorded the highest zinc content in green plant parts.
- Ukelp had lowest calcium content in necrotic part
- 5 ml/l SPB24P5 showed highest zinc content



Mineral Nutrient analysis of healthy part



- Significant differences in mineral concentrations
- AMLLK had the highest calcium content.
- SPAP4 showed the highest zinc and magnesium concentrations.
- The means were not significantly different ($p > 0.05$).



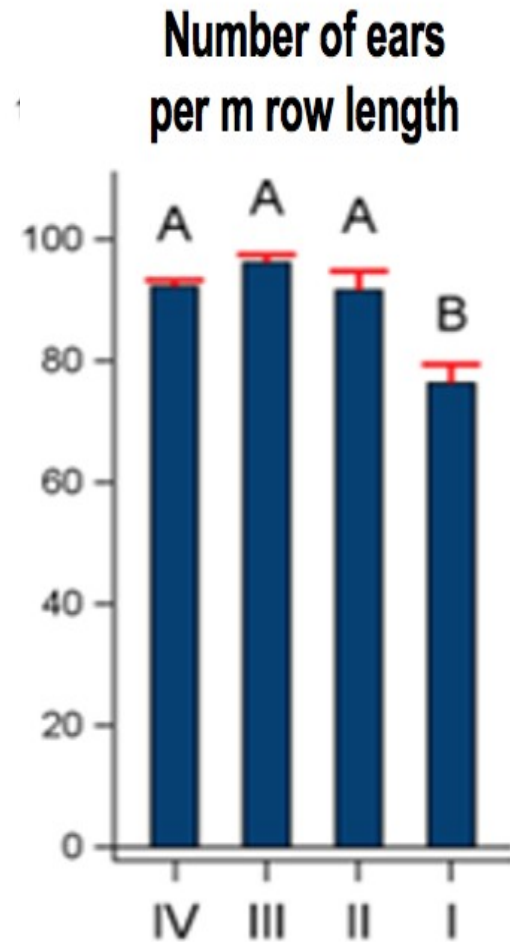
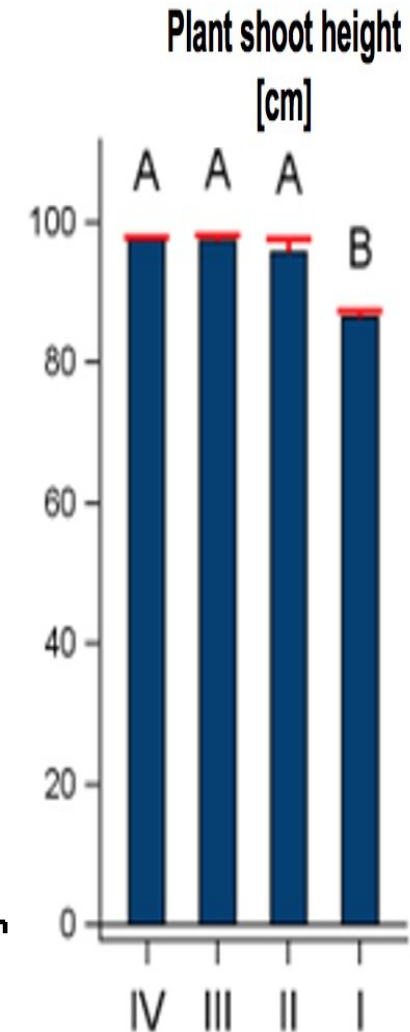
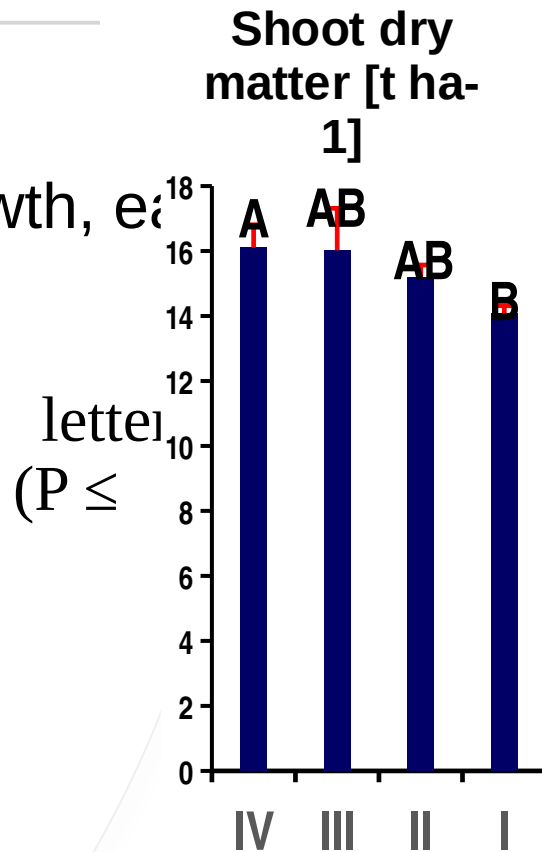
Potassium and phosphorus showed no significant differences.

SPAP4 had the highest Zinc content

Results of the field experiment at Gut Wilhelmstal

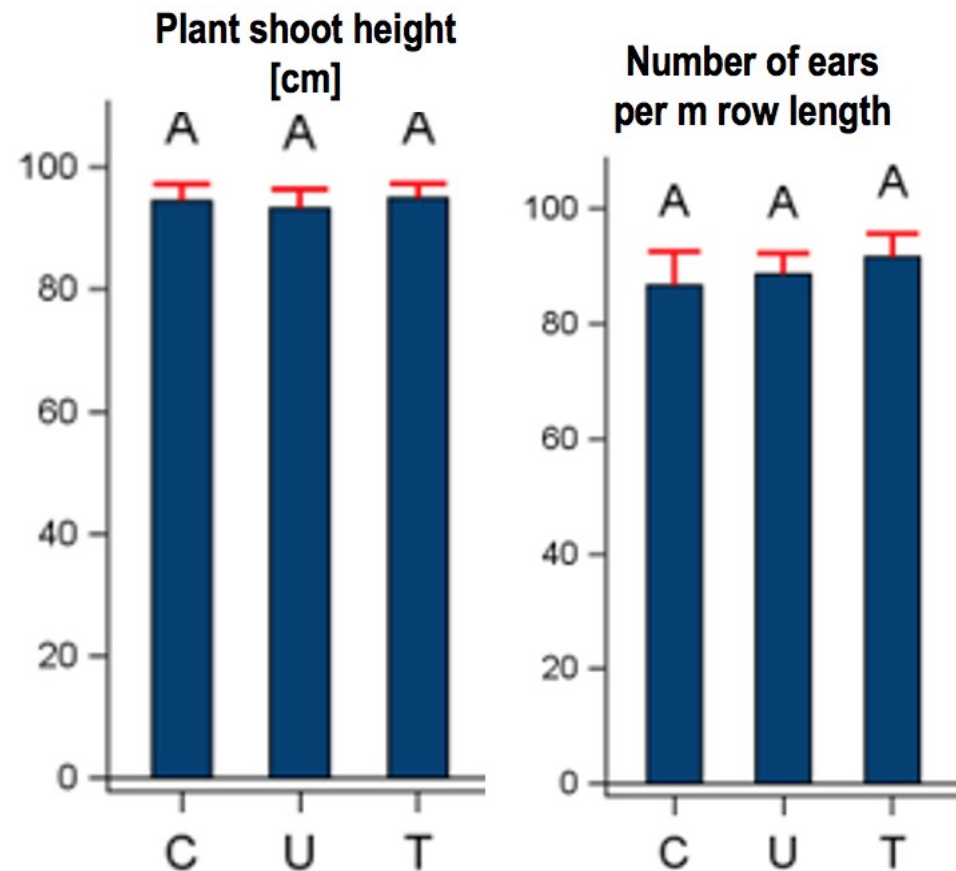
Plant growth

- Block effects significantly had influence.
- Block I exhibited the lowest growth, ear number and dry mass.
- Blocks or treatments sharing same letter are not significantly different ($P \leq 0.05$)

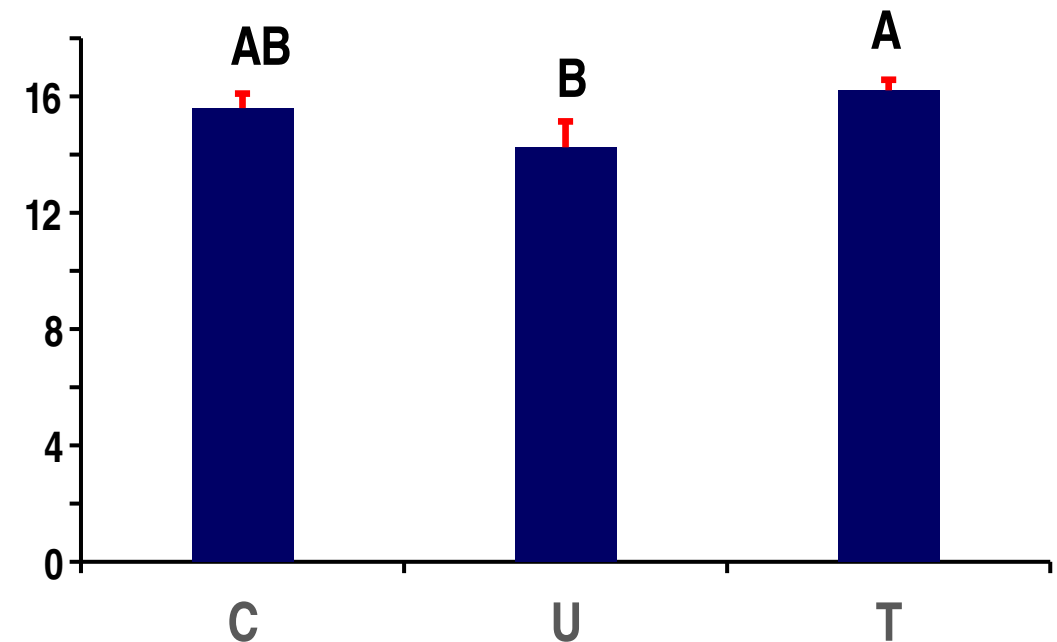


- Bio-effector treatments showed no significant differences in wheat growth.
- TopHealth produced the highest dry mass compared to Ukelp.
- There was a slight yield increase

	Grain yield	comparison to Control
Control	6.491 t/ha	100.0%
UKelp	6.670 t/ha	102.7%
TopHealth	6.668 t/ha	102.7%
MW	6.669 t/ha	

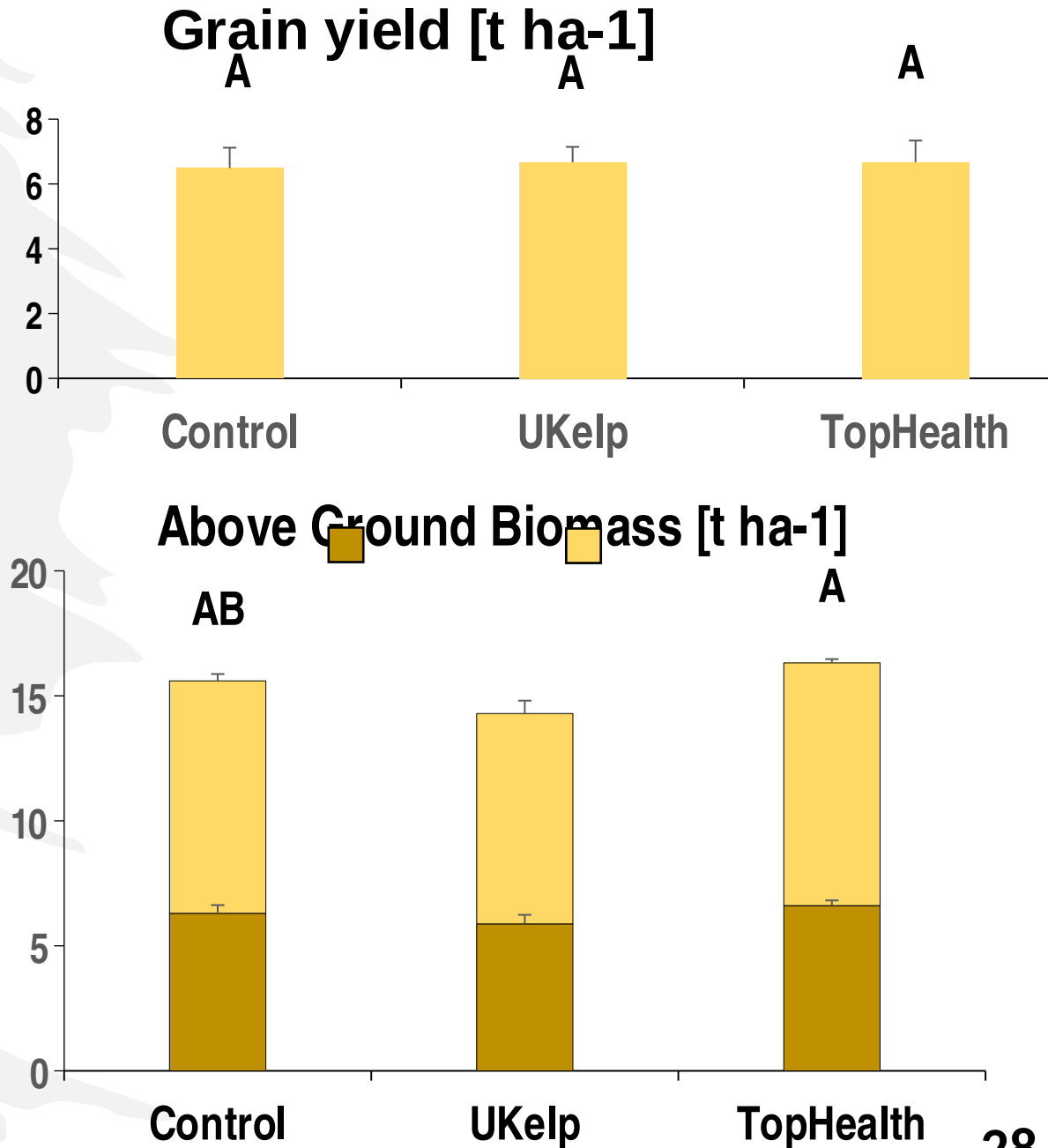


Shoot dry matter [t ha⁻¹]



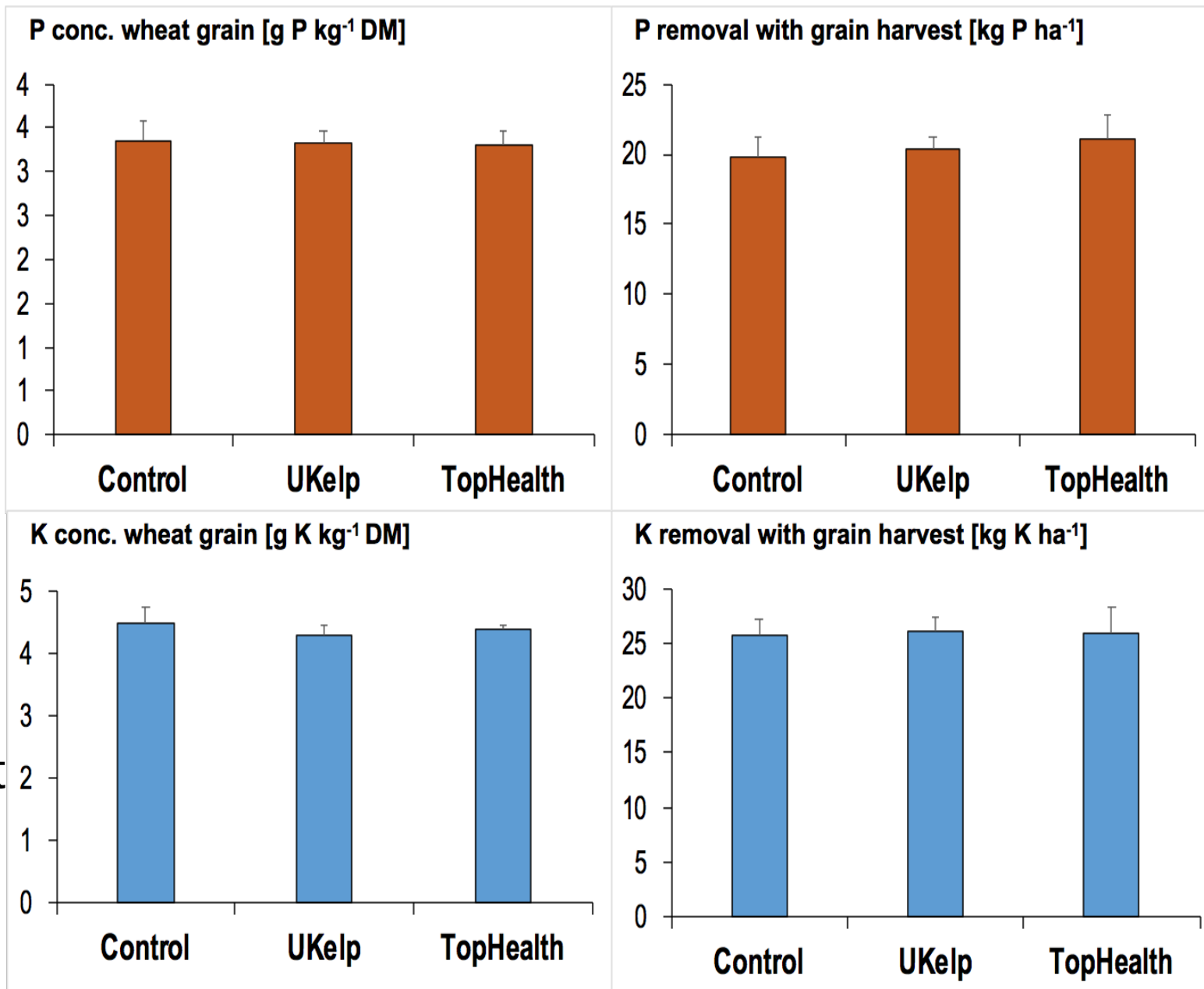
Grain yield and Biomass

- No significant difference in grain yield
- Ukelp & TopHealth had same grain yields.
- TopHealth had highest straw & ear biomass
- The control had similar biomass to Ukelp and TopHealth.
- Same letters= No significant difference



Field experiment Nutrient test results

- No significant differences in phosphorus and potassium concentrations.
- Phosphorus and potassium removal showed no significant differences.



Discussion

- Bio-effectors improved chlorophyll and water use efficiency.
- Plant height, biomass, and yield showed limited effects.
- Mineral accumulation showed little statistical significance in durum wheat.
- Innovative seaweed extracts showed potential for drought resilience and growth.
- Further optimization of bio-effector application strategies is necessary.
- There was a slight grain yield increase of winter wheat at the field

Discussion

- **Szczepanek et al. (2018)**: No significant difference in wheat growth with *Kelpak*.
- **Shibaeva et al. (2023)**: Higher seaweed concentrations reduce growth; lower concentrations improve it.
- **Al-Hasany et al. (2019)**: Seaweed extracts delay senescence, protect chloroplasts under drought.
- **Basavaraja et al. (2018)**: Seaweed extracts enhance uptake of N, P, K, and S in wheat and maize.
- **Rathore et al. (2008)**: High seaweed extract concentrations improve nutrient uptake in crops.
- **Chouliars et al. (2009)**: *E. maxima* and *A. nodosum* improve yields and quality in mustard and olive.

Conclusion



Seaweed extracts improved chlorophyll, & water use during drought phase.



No significant impact on plant height or yield.



Minimal effect on mineral accumulation in durum wheat.



Further research needed to optimize seaweed extract applications.

Thank You for Kind Attention

Jean Jules Cesar & Lokesh



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